

balanced panel and partial out unit and time fixed effects from y_{it} and D_{it} using the Frisch-Waugh theorem (Frisch and Waugh 1933). Denote grand means by $\bar{\bar{x}} = \frac{1}{NT} \sum_i \sum_t x_{it}$, and adjusted variables by $\tilde{x}_{it} = (x_{it} - \bar{\bar{x}}) - (\bar{x}_i - \bar{\bar{x}}) - (\bar{x}_t - \bar{\bar{x}})$. $\hat{\beta}^{DD}$ then equals the univariate regression coefficient between adjusted outcome and treatment variables:

$$\frac{\widehat{cov}(\tilde{y}_{it}, \tilde{D}_{it})}{\widehat{var}(\tilde{D}_{it})} = \frac{\frac{1}{NT} \sum_i \sum_t \tilde{y}_{it} \tilde{D}_{it}}{\frac{1}{NT} \sum_i \sum_t \tilde{D}_{it}^2}$$

The numerator equals the sample covariance between y_{it} and D_{it} minus the sample covariances between unit means and between time means:

$$\widehat{cov}(\tilde{y}_{it}, \tilde{D}_{it}) = \frac{1}{NT} \sum_i \sum_t (y_{it} - \bar{y}) (D_{it} - \bar{D}) - \frac{1}{N} \sum_i (\bar{y}_i - \bar{y}) (\bar{D}_i - \bar{D}) - \frac{1}{T} \sum_t (\bar{y}_t - \bar{y}) (\bar{D}_t - \bar{D}) \quad (6)$$

The appendix shows how to simplify this covariance using the binary nature of D_{it} , and by replacing the $\bar{y}_i$ and $\bar{y}_t$ terms with weighted averages of pre- and post-treatment means or means in each group.¹⁰ This gives the following theorem:

Theorem 1. Difference-in-Differences Decomposition Theorem

Assume that the data contain $k = 1, \dots, K$ groups of units ordered by the time when they receive a binary treatment, $t_k^* \in (1, T]$. There may be one group, U , that never receives treatment. The OLS estimate, $\hat{\beta}^{DD}$, in a two-way fixed-effects model (2) is a weighted average of all possible two-by-two DD estimators.

$$\hat{\beta}^{DD} = \sum_{k \neq U} s_{kU} \hat{\beta}_{kU}^{2x2} + \sum_{k \neq U} \sum_{\ell > k} s_{k\ell} [\mu_{k\ell} \hat{\beta}_{k\ell}^{2x2,k} + (1 - \mu_{k\ell}) \hat{\beta}_{k\ell}^{2x2,\ell}] \quad (7)$$

Where the two-by-two DD estimators are:

$$\hat{\beta}_{kU}^{2x2} \equiv (\bar{y}_k^{POST(k)} - \bar{y}_k^{PRE(k)}) - (\bar{y}_U^{POST(j)} - \bar{y}_U^{PRE(j)})$$

¹⁰ Note that the first two terms in equation (6) collapse to a function of pre/post mean differences because $D_{it} = \bar{D}_i = 0$ in the untreated group, $D_{it} = 0$ in pre-treatment periods, and $\bar{y}_i = \bar{D}_i \bar{y}_i^{POST(k)} - (1 - \bar{D}_i) \bar{y}_i^{PRE(k)}$. The sums over i thus include only the treated units and the sum over t includes only post-periods ($t > t_k^*$). Similarly, sum over time can be broken into pieces based on the different groups' treatment times, and the time means of y_{it} weight together each group's mean. $\widehat{var}(\tilde{D}_{it})$ follows from replacing y_{it} with D_{it} in the expression for $\widehat{cov}(\tilde{y}_{it}, \tilde{D}_{it})$.

$$\begin{aligned}\hat{\beta}_{k\ell}^{2x2,k} &\equiv \left(\bar{y}_k^{MID(k,\ell)} - \bar{y}_k^{PRE(k)}\right) - \left(\bar{y}_\ell^{MID(k,\ell)} - \bar{y}_\ell^{PRE(k)}\right) \\ \hat{\beta}_{k\ell}^{2x2,\ell} &\equiv \left(\bar{y}_\ell^{POST(\ell)} - \bar{y}_\ell^{MID(k,\ell)}\right) - \left(\bar{y}_k^{POST(\ell)} - \bar{y}_k^{MID(k,\ell)}\right)\end{aligned}$$

the weights are:

$$\begin{aligned}s_{kU} &= \frac{n_k n_U \bar{D}_k (1 - \bar{D}_k)}{\widehat{var}(\tilde{D}_{it})} \\ s_{k\ell} &= \frac{n_k n_\ell (\bar{D}_k - \bar{D}_\ell)(1 - (\bar{D}_k - \bar{D}_\ell))}{\widehat{var}(\tilde{D}_{it})} \\ \mu_{k\ell} &= \frac{1 - \bar{D}_k}{1 - (\bar{D}_k - \bar{D}_\ell)}\end{aligned}$$

and $\sum_{k \neq U} s_{kU} + \sum_{k \neq U} \sum_{\ell > k} s_{k\ell} = 1$.

Proof: See appendix A.

Theorem 1 completely describes the sources of identifying variation in a general DD model and their importance. With K timing groups, one could form $K^2 - K$ “timing-only” estimates that either compare an earlier- to a later-treated timing group ($\hat{\beta}_{k\ell}^{2x2,k}$) or a later- to earlier-treated timing group ($\hat{\beta}_{k\ell}^{2x2,\ell}$). With an untreated group, one could form K 2x2 DD’s that compare each timing group to the untreated group ($\hat{\beta}_{kU}^{2x2}$). Therefore, with K timing groups and one untreated group, the DD estimator comes from K^2 distinct 2x2 DDs.

The weights come both from group sizes, via the n_j ’s, and the treatment variance in each pair.¹¹ With one treated group, the variance of the treatment dummy is $\bar{D}_k(1 - \bar{D}_k)$, and is highest

¹¹ Many other least-squares estimators weight heterogeneity this way. A univariate regression coefficient equals an average of coefficients in mutually exclusive (and demeaned) subsamples weighted by size and the subsample x - variance:

$$\hat{\alpha} = \frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2} = \frac{\sum_A (y - \bar{y})(x - \bar{x}) + \sum_B (y - \bar{y})(x - \bar{x})}{\sum_i (x - \bar{x})^2} = \frac{n_A s_{xy}^A + n_B s_{xy}^B}{s_{xx}^2} = \frac{n_A s_{xx}^{2,A}}{s_{xx}^2} \hat{\alpha}_A + \frac{n_B s_{xx}^{2,B}}{s_{xx}^2} \hat{\alpha}_B$$

Similarly, the Wald/IV theorem (Angrist 1988) shows that any IV estimate is a linear combination of Wald estimators that compare two values of the instrument. Gibbons, Serrato, and Urbancic (2018) show a nearly identical weighting formula for one-way fixed effects. Panel data provide another well-known example: a pooled regression coefficients equals a variance-weighted average of two distinct estimators that each use less information: the between estimator for subsample means, and the within estimator for deviations from subsample means.

for units treated in the middle of the panel with $\bar{D}_k = 0.5$. With two treated groups, the variance of the difference in treatment dummies is $(\bar{D}_k - \bar{D}_\ell)(1 - (\bar{D}_k - \bar{D}_\ell))$, and is highest for pairs whose treatment shares *differ* by 0.5. Figure 2 sets t_k^* and t_ℓ^* so that $\bar{D}_k = 0.67$ and $\bar{D}_\ell = 0.15$, which means that $\bar{D}_k(1 - \bar{D}_k) = 0.22 > 0.13 = \bar{D}_\ell(1 - \bar{D}_\ell)$. Because it has higher treatment variance, group k 's comparison to the untreated group, $\hat{\beta}_{kU}^{2 \times 2}$, gets more weight ($s_{kU} = 0.365$) than the corresponding term for group ℓ , $\hat{\beta}_{\ell U}^{2 \times 2}$ ($s_{\ell U} = 0.202$). The difference in treated shares is $\bar{D}_k - \bar{D}_\ell = 0.52$, so $s_{k\ell} = 0.412$.¹²

This decomposition theorem also shows how DD compares groups treated at different times. A two-group “timing-only” estimator is itself a weighted average of the 2x2 DD's plotted in panels C and D of figure 2:

$$\hat{\beta}_{k\ell}^{2 \times 2} \equiv \mu_{k\ell} \hat{\beta}_{k\ell}^{2 \times 2, k} + (1 - \mu_{k\ell}) \hat{\beta}_{k\ell}^{2 \times 2, \ell} \quad (8)$$

Both groups serve as controls for each other during periods when their treatment status does not change, and the group with higher treatment variance (that is, treated closer to the middle the panel) gets more weight. In the three group example $\bar{D}_k(1 - \bar{D}_k) > \bar{D}_\ell(1 - \bar{D}_\ell)$ so $\mu_{k\ell} = .67$. Multiplying this by $s_{k\ell} = 0.412$, shows that $\hat{\beta}_{k\ell}^{2 \times 2, \ell}$ gets less weight than $\hat{\beta}_{k\ell}^{2 \times 2, k}$: $s_{k\ell}(1 - \mu_{k\ell}) = 0.135 < 0.278 = s_{k\ell}\mu_{k\ell}$.¹³

¹² Changing the number or spacing of time periods changes the weights. Imagine adding T periods to the end of the three-group panel. This would reduce $\widehat{var}(D_{kt})$ to $0.835(1 - 0.835) = 0.138$, but it would increase $\widehat{var}(D_{\ell t})$ to $0.58(1 - 0.58) = 0.244$. This puts less weight on terms where k is the treatment group ($s_{kU} = 0.24$ and $s_{k\ell}\mu_{k\ell} = 0.07$) and more weight on terms where ℓ is the treatment group ($s_{\ell U} = 0.43$ and $s_{k\ell}(1 - \mu_{k\ell}) = 0.26$).

¹³ Two recent papers present clear analyses using two-group timing-only estimators. Malkova (2017) studies a maternity benefit policy in the Soviet Union and Goodman (2017) studies high school math mandates. Both papers show differences between early and late groups before the reform, $PRE(k)$, during the period when treatment status differs, $MID(k, \ell)$, and in the period after both have implemented reforms, $POST(\ell)$.

II. THEORY: WHAT PARAMETER DOES DD IDENTIFY AND UNDER WHAT ASSUMPTIONS?

Theorem 1 relates the regression DD coefficient to sample averages, which makes it simple to analyze its statistical properties by writing $\hat{\beta}^{DD}$ in terms of potential outcomes (Holland 1986, Rubin 1974). The outcome is $y_{it} = D_{it}Y_{it}^1 + (1 - D_{it})Y_{it}^0$, where Y_{it}^1 is unit i 's treated outcome at time t , and Y_{it}^0 is the corresponding untreated outcome. Following Callaway and Sant'Anna (2018, p 7) define the ATT for timing group k at time τ (the “group-time average treatment effect”): $ATT_k(\tau) \equiv E[Y_{it}^1 - Y_{it}^0 | k, t = \tau]$. Because regression DD averages outcomes in pre- and post-periods, I define the average of the $ATT_k(\tau)$ in a date range, W :

$$ATT_k(W) \equiv E[Y_{it}^1 - Y_{it}^0 | k, t \in W] \quad (9)$$

In practice, W will represent post-treatment windows that appear in the 2x2 components. Finally, define the difference over time in average potential outcomes (treated or untreated) as:

$$\Delta Y_k^h(W_1, W_0) \equiv E[Y_{it}^h | k, W_1] - E[Y_{it}^h | k, W_0], \quad h = 0, 1 \quad (10)$$

Applying this notation to the 2x2 DD's in equations (3)-(5), adding and subtracting post-period counterfactual outcomes for the treatment group yields the familiar result that (the probability limit of) each 2x2 DD equals an ATT plus bias from differential trends:

$$\beta_{kU}^{2x2} = ATT_k(POST(k)) + \Delta Y_k^0(POST(k), PRE(k)) - \Delta Y_U^0(POST(k), PRE(k)) \quad (11a)$$

$$\beta_{k\ell}^{2x2,k} = ATT_k(MID(k, \ell)) + \Delta Y_k^0(MID(k, \ell), PRE(k)) - \Delta Y_\ell^0(MID(k, \ell), PRE(k)) \quad (11b)$$

$$\begin{aligned} \beta_{k\ell}^{2x2,\ell} &= ATT_\ell(POST(\ell)) + \Delta Y_\ell^0(POST(\ell), MID(k, \ell)) - \Delta Y_k^0(POST(\ell), MID(k, \ell)) \\ &\quad - [ATT_k(POST(\ell)) - ATT_k(MID(k, \ell))] \end{aligned} \quad (11c)$$

Note that the definition of common trends in (11a) and (11b) involves only counterfactual outcomes, but in (11c) identification of $ATT_\ell(POST(\ell))$ involves counterfactual outcomes *and* changes in treatment effects in the already-treated “control group”.